

Experiment 1

1. Objective

Get familiar with the Linux operating system environment through experiments and master the use of basic Linux commands.

2. Requirement

- Master the permission management of Linux files
- Master basic file and directory operations
- Linux user management
- File compression and archiving

3. Experiment Step

(1) Directory Operation

Enter the following commands in sequence:

```
$pwd
$cd ~
$mkdir last_4_numbers_of_your_studentID
$cd last_4_numbers_of_your_studentID
$mkdir cs
$ls
$rmkdir cs
$ls
```

Save a screenshot of the above command along with the corresponding results, you must have your student number and name.

(2) File operation

Enter the following commands in sequence:

```
$cd ~
$touch yourname.txt
$ls
$echo "hello world" > last_4_numbers_of_your_studentID.txt
$cat last_4_number_of_your_studentID.txt
$cp yourname.txt last_4_numbers_of_your_studentID_copy.txt
$mv yourname.txt last_4_numbers_of_your_studentID_newname.txt
$ls
```

Save a screenshot of the above command along with the corresponding results, you must have your student number and name.

(3)User Management

Note: You have to use the last 4 numbers of your student id to replace 0101 in the following.

Enter the following commands in sequence:

```
$whoami
$sudo su root
```

Enter the password of the current logged-in user, not the root password, and then enter:

```
#groupadd student
#whoami
#adduser NY0101
#usermod -g student NY0101
```

Follow the prompts to create the user and add the user to the student group.

Switch to the user.

```
#su NY0101
$cd ~
$touch NY0101.txt
$ls -l
```

View the properties of the file, including the owner user and the group to which the owner user belongs.

(4) Permission management

Enter the following commands in sequence:

```

$ touch demo
$ chmod 777 demo
$ chmod a-x demo          # result is rw-rw-rw-
$ chmod go-w demo         # result is rw-r--r--
$ chmod g+w demo          # result is rw-rw-r--
$ chmod a= demo           # result is -----
$ chmod +t demo           # result is -----T
$ chmod u+s demo          # result is --S-----T
$ chmod u+rx demo         # result is r-s-----T
$ chmod g+rws demo        # result is r-srwS--T

```

After entering each line, run `ls -l demo` to view the result.

###(5) Other management

Create folder "others" in the main directory of the user, go to the folder, and create two files with the name of any file. Then run the following commands in sequence.

```

$tar -czvf backup.tar.gz others
$ls -l
$tar -xzvf backup.tar.gz
$ls -l

```

Observe changes in the process.

Create a textfile file and make it as follows:

```

no, I study in njupt.
this line ends with 00
this line ends with .00
I am 25 years old.
He is 15. It is 16.00
wy is the short of my name.

```

Search for all lines beginning with the character "n" in the file textfile

```
$ grep '^n' textfile
```

Search the textfile for all lines ending in ".00"

```
$ grep '\.00$' textfile
```

Search for all lines in the textfile that contain the number 5, followed by the character ".", followed by any character

```
$ grep '5\..' textfile
```

Search for all lines starting with the characters "w" and "y" in the file textfile

```
$ grep '^[wy]'textfile
```